

We saw that b^x was either increasing or decreasing for $b \neq 1$, so it is one-to-one and thus has an inverse.

$$\text{If } f(x) = b^x, f^{-1}(x) = \log_b x$$

$$\log_b x = y \iff b^y = x \quad \log_b(b^x) = x \text{ for all } x$$

$$b^{\log_b(x)} = x \text{ for } x > 0$$

Ex

$$\log_3 81 = 4$$
$$\log_{25} 5 = \frac{1}{2}$$
$$\log_{10} 0.001 = -3$$

$f(x) = b^x$ has domain $\mathbb{R}$ and range $(0, \infty)$
So

$f^{-1}(x) = \log_b x$ has domain $(0, \infty)$ and
range $\mathbb{R}$.

Properties of logs

If $b > 1$, $\log_b(x)$ is one-to-one, increasing
w/ domain $(0, \infty)$, range $\mathbb{R}$. If $x, y > 0$, $r \in \mathbb{R}$
then

- $\log_b(xy) = \log_b(x) + \log_b(y)$
- $\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$
- $\log_b(x^r) = r \log_b(x)$

$$\lim_{x \rightarrow 0^+} \log_b(x) = -\infty, \quad \lim_{x \rightarrow \infty} \log_b(x) = \infty$$

Ex $\lim_{t \rightarrow 0^+} \log_{10}(t \cdot t^2) = \lim_{t \rightarrow 0^+} \log_{10}(t) = -\infty$

Generally, we only deal with
 $\log_e x = \ln x$

a nice property is any power or exponent

$$x^r = e^{\ln x^r} = e^{r \ln x} \quad x > 0$$

$$b^x = e^{\ln b^x} = e^{x \ln b}, \quad b > 0$$

$$\text{So, } \frac{d}{dx}(b^x) = \frac{d}{dx} e^{x \ln b} = \ln b \cdot e^{x \ln b} = \ln(b) b^x$$

Ex Solve $e^{5-3x} = 10$

$$5 - 3x = \ln(10)$$

$$x = \frac{5 - \ln(10)}{3}$$

Ex Rewrite $\ln\left(\frac{x^2\sqrt{x^2+2}}{3x+1}\right)$

$$= \ln(x^2) + \frac{1}{2}\ln(x^2+2) - \ln(3x+1)$$